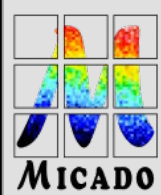




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MICADO

Entanglement of HW & SW

AO4ASTRO, Hugo Buddelmeijer, 2019/03/27

w/ Gijs Verdoes Kleijn, Eline Tolstoy

https://docs.google.com/presentation/d/1XtS_8UaVhsDNkwwU7aXMqGmc5gTserXmVYRfqyhCDws

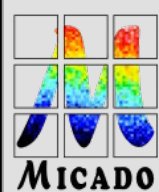
<http://bit.do/HB4AO4ASTRO>



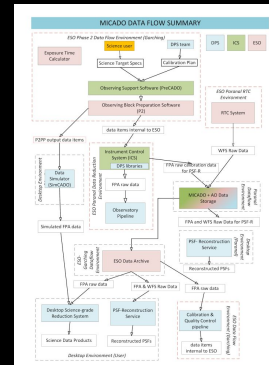
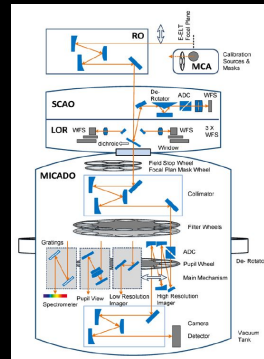
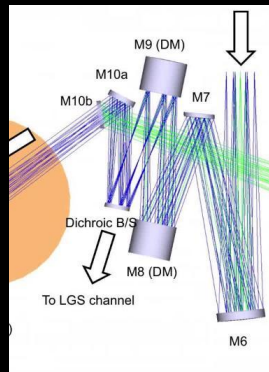
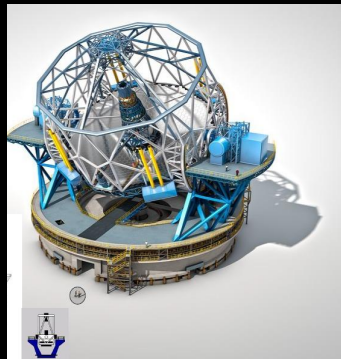
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OCA + ELT + MAORY + MICADO + DFS



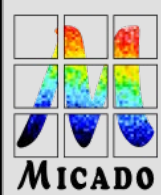
Observatory + Telescope + AO + Instrument + Processing



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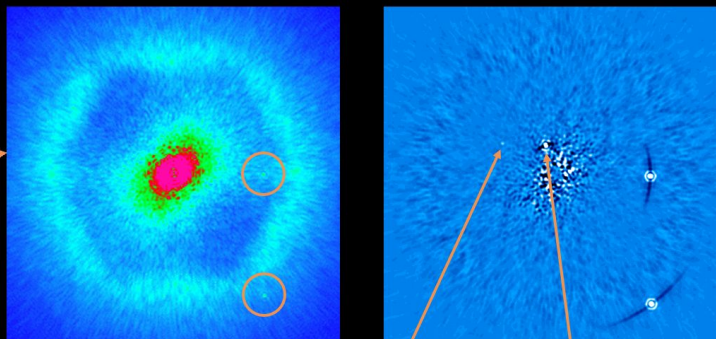
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Science Cases require HW + SW entanglement

High Contrast Imaging

MICADO simulation



added exoplanet at 10 AU, 700 K, $\log(g)=4$.

added exoplanet at 5 AU, 1300 K, $\log(g)=4$.

Precision Astrometry (see Davide Massari)

PSF-reconstruction (see Roland Wagner)

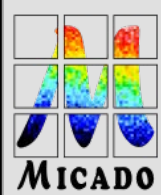
Persistence



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Traditional Situation

Hierarchical

1. Design Perfect Hardware.
2. Design Perfect Software.
3. Oops, can't do PSF-reconstruction properly.

ELT Situation

More Holistic, more trade-offs

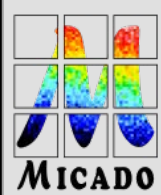
- Design best hardware and best software together.
- Oops, organisational entanglement skyrockets.



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Hugo: WP Imaging Lead

Main Responsibility:

- Design Imaging SW (to deliver as pdf's)

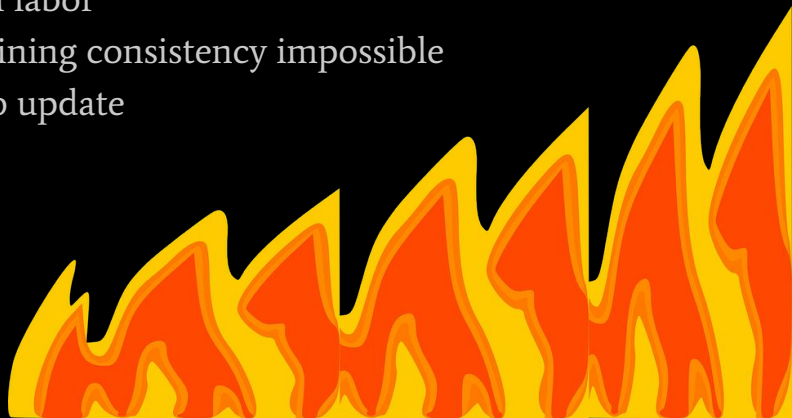
But then:

- Design Collaboration (SPEC, PSF-R, AO, ...)
- Algorithm Prototypes
- Simulator Input
- Archive
- ESO Data Handling System
- Consortium Data Handling System

Everything needs this design!

Legacy document-based process:

- Manual labor
- Maintaining consistency impossible
- Hard to update
- -> Hell

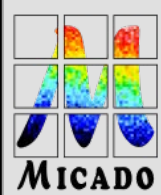




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Better: Digital Design

- The ‘What’:
 - Software Readable Design (XML or so)
- + ‘Why’:
 - Design Documents (Sphinx or so)
- + ‘How’:
 - Simulations (Python)
 - ESO implementation (C)
 - Consortium implementation (Python)
- Fully automatic:
 - Archive (SQL)

Benefits

- Design always internally consistent
- Changes propagate automatically
- Collaboration without hassle
- -> Heaven

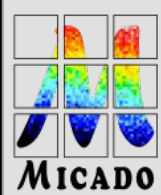




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MasterDark Example: “What” in XML

- There is a thing called “Master Dark”
- Needed to meet Requirement XYZ
- It is an image
- Created from “Raw Darks”
- Which can be found in this way
- Using recipe such-and-such
- Parameter A, B and C control its quality

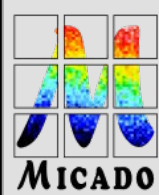
```
1028 <!-- Master Dark frame -->
1029 <xs:complexType name="nirMasterDarkFrame">
1030   <xs:annotation>
1031     <xs:documentation>Master dark reduced frame</xs:documentation>
1032   </xs:annotation>
1033   <xs:complexContent>
1034     <xs:extension base="baseCalibImageFrame">
1035       <xs:sequence>
1036         <xs:element name="ReadoutMode" type="leInisp:nispReadoutMode" minOccurs="0">
1037           <xs:annotation>
1038             <xs:documentation>Readout Method for Darks acquisition</xs:documentation>
1039           </xs:annotation>
1040         </xs:element>
1041         <xs:element name="Filter" type="ins:baseFilter">
1042           <xs:annotation>
1043             <xs:documentation>Instrument filter name</xs:documentation>
1044           </xs:annotation>
1045         </xs:element>
1046         <xs:element name="MSDParameters" type="msdDarkParameters">
1047           <xs:annotation>
1048             <xs:documentation>MasterDark Parameters</xs:documentation>
1049           </xs:annotation>
1050         </xs:element>
1051         <xs:element name="DataStorage" type="masterDarkFitsFile">
1052           <xs:annotation>
1053             <xs:documentation>MasterDark fits file
1054             </xs:documentation>
1055           </xs:annotation>
1056         </xs:element>
1057         <xs:element name="DarkFrames" type="darkFrameRefList" minOccurs="0">
1058           <xs:annotation>
1059             <xs:documentation>List of dark frames used for
1060             MasterDark creation</xs:documentation>
1061           </xs:annotation>
1062         </xs:element>
1063         <xs:element name="QualityParams" type="nirDocDetectorImageList" minOccurs="0">
```



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MasterDark Example: “Why” in RST -> PDF, HTML, ...

- A special procedure is needed to requirement XYZ because ...
- See this nice Figure ...

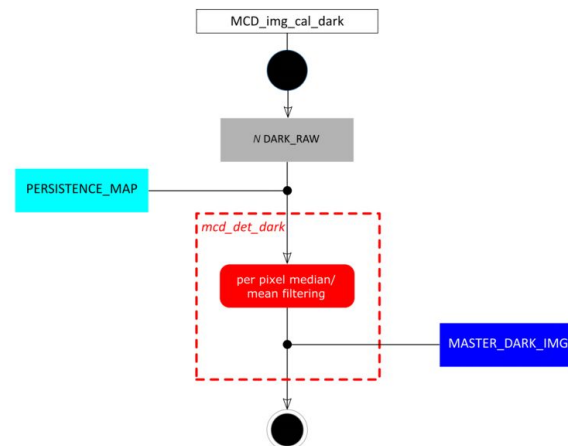
7.2.1 Dark

Purpose:

Darks will be used for the dark current correction and requires the same exposure time as the science data. Thus, these frames have to be taken on daily basis, but during day time.

Algorithm and workflow:

The observation template `MCD_img_cal_dark` provides N dark frames, which are median or mean combined to create the `MASTER_DARK_IMG` frame (see Fig.7.1.2). The input frames have to be taken with the same exposure time as the `SCIENCE` exposures. See Fig. 7.1.2 for the preliminary workflow.

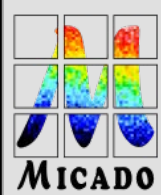




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MasterDark Example: “How” in Python / C

- The Master Dark is created by median stacking the Raw Darks

```
escape = None

darks: [Dark] = persistent("A list of raw darks [None]", Dark,
[1])

def make(self):
    """
    Process a MasterDark.
    """
    images = [dark.image for dark in self.darks]
    self.image = numpy.median(images, axis=0)
```

TODO Python Console Terminal 9: Version Control Event Log

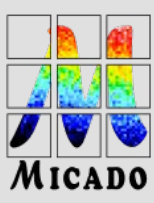
kerfile detection // You may s... (7 minutes ago) 11:1 LF UTF-8 Git: 9-include-python-cpl



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Take-away how to deal w/ entangled SW/HW design

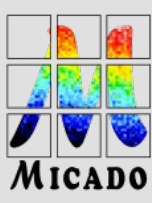
- Software and Hardware design will be more and more entangled.
- Seek collaboration with your Hardware / Software designer peer.
- Use, share and re-use with machine readable formats.



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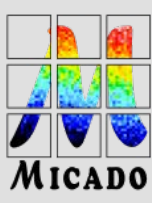




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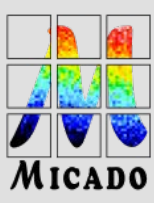




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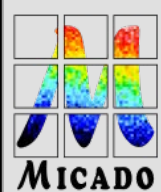
Extra Slides



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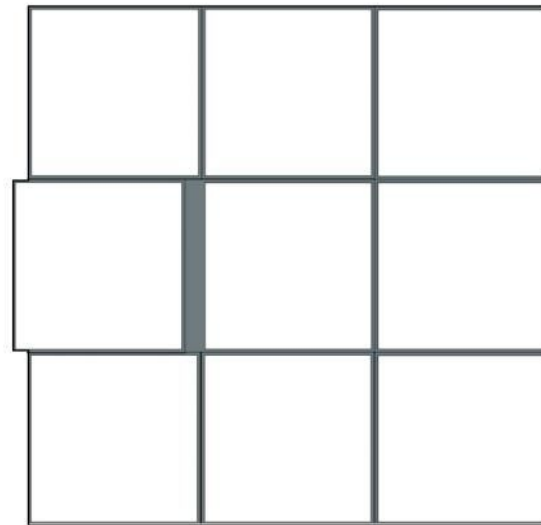


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Role HB: WP Imaging Lead

- Pipeline Design
- Combine Astro-WISE + ESO tech
- Backward Chaining Data Model
- Prototypes Critical Algorithms
 - Astrometry
- Processing Simulations
- Etc.



189mm, 50.5" (4mas/pix), 19" (1.5mas/pix)

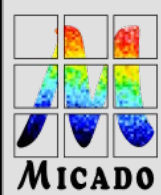
9 4K by 4K H4RG detectors
2019/03/27 AO4ASTRO hugo@buddelmeijer.nl



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ELT + MICADO @Kapteyn: Hardware + Software $\Leftrightarrow$ Astronomy

Astronomy Science

- Eline Tolstoy (Kapteyn)
- Helmi, Massari, Caputi, Dayal, Morganti, Koopmans, Trager, +

Data Science

- Gijs Verdoes Kleijn (Kapteyn, OmegaCEN)
- Hugo Buddelmeijer (Kapteyn, OmegaCEN)
- + OmegaCEN team

Instrument Science

- Ramon Navarro (NOVA-ASTRON)
- Annemieke Janssen (NOVA-ASTRON)
- Joost van den Born (RUG PhD @NOVA-ASTRON)
- + NOVA-ASTRON team
- + Leiden High Contrast Imaging team

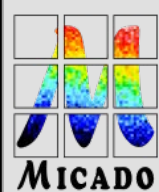




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MasterDark Example

Fig. 7.1.2: Workflow diagram for the masterdark creation recipe

Name	Dark current
ReqID	DPS-3000-SCI
Parent ReqID	[R-INS-741]D // [R-INS-908]D //
Objective	Determine the dark current.
Responsible	Science Operations
Phase	Day time
MICADO Set-up	* Ensure calibration Unit off * Set wheel with blocking filter in right position * Make exposures
Calibration Unit setup	Not active
Template name	MCD_img_cal_dark
Input parameters	DIT,NDIT,NEXPO,READ_OUT_MODE,WINDOW,DPR,CATG,EXPOSURE_TIME,EXPOSURE_NUMBER
Observing Block name	To be added at FDR.
Observations frequency	Daily (8C).
Observations duration	476sec
Prerequisites	None
Recipe frequency	Once per Observing Block with dark frames.
Input products	DARK_RAW,PERSISTENCE_IMG
Output products	master dark image (MASTERDARK_IMG) QC DARKMEAN,QC DARKMED,QC DARKRMS,QC NPXSAT,QC FPIXSAT

```
micadowise - [C:\Users\Hugo\hugo\micado\repos\micadowise] - _src\micado\main...
File Edit View Navigate Code Refactor Run Tools VCS Window Help

micadowise src micado main
MasterDark.py
8 from common.database.DBMain import persistent
9 from micado.main.BaseFrame import BaseFrame
10 from micado.main.Dark import Dark
11
12
13 class MasterDark(BaseFrame):
14     """
15     A master dark file.
16
17     See MASTERDARK_IMG DataTable:
18     https://docs.google.com/spreadsheets/d
19     /1BbQkF818cCFStf7akEBy0Gc2UMK9kPWBkY0o3hZqVs/edit#gid=751586500
20     &range=4:4
21
22     See mod_det_dark recipe:
23     https://docs.google.com/spreadsheets/d
24     /1BbQkF818cCFStf7akEBy0Gc2UMK9kPWBkY0o3hZqVs/edit#gid=1072399737
25     &range=5:5
26     https://docs.google.com/spreadsheets/d
27     /1t1OGqcxWQo3TeUQcN9GsbM6b4ful4xTLAdQr2Ug/edit#gid=694297875
28     """
29
30     recipe = "mod_det_dark"
31
32     darks: [Dark] = persistent("A list of raw darks (None)", Dark,
33     [])
34
35     def make(self):
36         """
37         Process a MasterDark.
38         """
39         images = [dark.image for dark in self.darks]
40         self.image = numpy.median(images, axis=0)
```

.xml: There is a Dark, ...

.rst: ... because of reasons, ...

py: ... created like this,
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